

Darb — Jordan Trips, Reality-Checked

Team PixelsDev

1. Project Overview

Idea Summary

Darb (Arabic for "path") is a web platform that helps travelers plan a trip to Jordan and checks that the plan works on the ground. Generic planners produce plans that look fine on paper; many of them cannot be carried out.

Users paste a ready-made itinerary (from ChatGPT, a blog or a travel agent) or build a new plan by answering a few questions. Darb then runs a **Reality Check**. It verifies that transport exists between cities at the needed time, that distances and timings are realistic, that sites are open, and whether a Jordan Pass saves money. Every issue comes with an alternative and its cost in Jordanian Dinars (JOD). The traveler gets a corrected plan they can follow and add to their phone calendar in one step, with packing advice for the season.

Travelers arrive in Jordan with a tested, fully costed plan. They do not find out on the ground that the only direct bus to Petra leaves at 06:30, that Wadi Rum cannot be booked on JETT at all, or that Jerash and the Dead Sea do not fit into one day. The same data shows the Ministry of Tourism where independent travelers want to go but cannot.

Target Users and Context of Use

Primary user: the first-time international visitor to Jordan, especially those who will not rent a car: solo travelers, students, couples and mid-budget travelers. In 2025, tourism revenue from European markets grew 39.8% and from Asian markets 32.7% [1]; most visitors from these markets are unfamiliar with Jordan's local transport network.

Their needs: how to get between cities and what it costs, whether the Jordan Pass is worth it, how many days they really need, and which places can be combined in one day.

Context of use: weeks before the trip, on a laptop or phone, while preparing their plan and comparing it with suggestions from ChatGPT or blogs. During the trip they return to the plan on their phone, so it must be saved and available offline.

Secondary users: the Ministry of Tourism and Antiquities and the Jordan Tourism Board (a demand and issues dashboard), and data owners such as JETT, the Petra Development and Tourism Region Authority (PDTRA) and nature reserves (updating their own schedules, prices and opening hours). Hostels and tour organizers follow in later phases.

Goals and Expected Outcomes

Goal 1 (executable plans): 90% of plans produced by Darb contain no transport or timing conflicts. Indicator: share of plans passing the Reality Check after correction, verified manually on a monthly sample.

Goal 2 (planning speed): users get a corrected, costed plan in under 5 minutes after pasting their itinerary. Indicator: time from first input to saving the plan.

Goal 3 (tangible savings): show each user the expected savings (Jordan Pass or cheaper transport), targeting an average of 25 JOD or more per trip. Indicator: average savings displayed by the platform.

Goal 4 (early adoption): 1,200 checked plans and at least 80 paid trips per month (operating break-even) within 6 months of launch, with a satisfaction score of 4/5 or higher in the post-trip survey.

2. Problem and Proposed Solution

Problem Definition and Impact

The problem: information about moving between tourist sites in Jordan is fragmented and contradictory, and generic planning tools, including AI assistants, produce itineraries that are not grounded in this reality.

Evidence:

- The World Bank's diagnostic of public transport in Jordan describes the service as having poor coverage, low frequencies and no integration of services and fares [2].
- **The operator contradicts the guides.** JETT's own booking system, checked on 24/09/2026 for 25/09/2026, shows one departure from Abdali to Petra at 06:30, arriving 10:30 (4 h, 10 JOD, 1 seat left). Wadi Rum does not appear as a bookable destination, even though JETT's homepage lists it [14]. Reaching Wadi Rum from Amman without a car costs about 110 JOD by taxi, or takes up to 8 hours by bus via Aqaba [4].
- **Third-party sources disagree with the operator and with each other:** a 2026 guide gives 11 JOD for Amman-Petra [3] and an aggregator USD 31 [19]; one guide says JETT has no Dead Sea service [3], another describes a daily 8:30 bus for about 15 JOD [5]; forum users report that schedules open only 1-2 months ahead [19].

This is the core evidence for Darb. Even we, checking four sources, got the price wrong until we opened the operator's booking system. A traveler has no chance.

The screenshot shows a web interface for JETT's booking system. At the top, there's a header with 'وقت المغادرة' (Departure Time) on the left and '1 بالغين البتراء ~ العبدلي' (1 Adult to Petra ~ Abdali) on the right. Below the header, there's a date selector with four options: '27-09-2026', '26-09-2026', '25-09-2026' (which is highlighted), and '24-09-2026'. Underneath the date selector, there's a section for the route and time. It shows 'العبدلي 06:30' (Abdali 06:30) and 'البتراء 10:30' (Petra 10:30) with a double-headed arrow between them. To the right of this, it says 'المدة: 4س 0د' (Duration: 4h 0m). Further right, there's a blue button labeled 'اختار' (Choose). At the bottom right, it displays 'JOD 10' and '1 مقعد متبقية' (1 seat remaining).

Figure A — JETT booking system, checked 24/09/2026: Amman (Abdali) → Petra 06:30–10:30, 10 JOD.

- A Greetwell survey of 1,000 U.S. travelers found that 55% of those who used AI for trip planning encountered at least one inaccurate recommendation, and 1 in 6 were recommended a place or activity that does not exist [6].
- **The segment Darb serves is the one that collapsed.** Petra received 1.17 million visitors in 2023 but only 457,215 in 2024. In January–July 2025 it received about 291,000,

of whom 193,061 (66%) were foreign and 78,889 Jordanian. In Petra, 32 hotels (2,000 rooms) closed and about 700 jobs were lost [15].

- **Visitors concentrate on a few sites.** Lesser-visited sites have room to grow: Umm Qais received 111,519 visitors in January–November 2025 (+47.4% year on year), but only 22,519 were foreign; Ajloun Forest Reserve received about 143,500 by October 2025 [16]. Foreign travelers rarely reach these places, partly because they cannot tell whether a site fits a trip without a car.

Impact: lost time and money, cancelled or compressed days, expensive unplanned taxi rides, and a negative impression of how easy Jordan is to travel. The sector generated USD 7.79 billion in revenue in 2025 [1], but much of that growth comes from Arab and GCC visitors, while Petra's foreign-heavy numbers fell [15]. Winning back the independent foreign traveler, and spreading them beyond Petra, is where reliable planning has the greatest economic effect.

Current Alternatives and the Gap

The team analyzed 35 competing websites and apps, 9 of them in depth, using a 15-feature matrix and a standard persona scenario (Sarah) with 12 tasks. The main alternatives:

- **ChatGPT and general AI assistants:** produce a neat itinerary quickly, but do not verify transport or timings and can recommend places and details that are wrong [6].

- **Mindtrip:** chat with a map and a pleasant experience, but its data is generic and does not cover Jordan's local transport details.

- **Rome2Rio:** shows transport options and approximate costs between two points, but does not build a full plan or check each day as a whole.

- **Official websites, blogs and TripAdvisor forums:** sometimes accurate, but scattered and possibly outdated or contradictory [3][5].

- **Visit Jordan (official app and website, Jordan Tourism Board):** strong on discovery (an interactive map of landmarks including lesser-known sites, a voice assistant, visa and transport tips, ticket storage), but adoption is minimal (500+ Android downloads) and neither the app nor the official website offers an itinerary planner or feasibility check [17]. Darb complements it and could be embedded as the official app's planning engine.

- **Tour operators and private drivers:** solve the problem at a high cost (e.g., about 144 JOD for a private driver day trip to Wadi Rum [4]) and do not suit independent travelers.

The gap: no existing tool takes a plan, checks it against Jordan's reality (transport, timings, prices, Jordan Pass) and fixes it automatically.

Solution and Added Value

Darb runs as **three connected loops**. Travelers produce plans, the ministry sees demand, and data owners keep the facts current.

Loop 1 (traveler): check and fix, or build a feasible plan.

1. **Input:** the user pastes a text itinerary, or chooses "Build a plan" and picks interests, pace, number of days and car / no car.
2. **Parsing:** a rule-based Plan Parser handles "Day N" text first; anything else goes to a model-agnostic AI Parsing Service that turns free text into structured days and places matched against the destination table. The user confirms the parsed days before the check.
3. **Checking:** a rules engine checks each day against verified, dated data: bus schedules, real travel times, opening hours, ticket prices and Jordan Pass rules.
4. **Result:** a Reality Score and color-coded issues (red: not feasible, yellow: risky, green: OK), each with an alternative and its cost. In "Build a plan", destination cards are labeled "fits your trip" or "needs +1 day or a car" before anything is added.
5. **Fix and save:** one click produces a corrected plan with every transport leg and the total cost, saved as a link or offline PDF. The offline plan includes a pre-computed "if you're late" alternative for every transport leg, and the traveler can add it to their calendar (.ics file or Google Calendar link): one event per day and per transport leg, with a reminder 45 minutes before each departure.

- **Weather & what to pack:** for each destination and travel month, the plan shows expected day and night temperatures and site-specific tips (e.g., walking shoes and 2 L of water for Petra; a warm jacket for Wadi Rum nights, even in October). When the trip starts within 7 days, a live forecast replaces the seasonal averages. The Reality Engine also applies seasonal timing rules: in summer Petra starts at opening time and avoids midday; in winter shorter daylight means fewer sites per day.

Loop 2 (ministry): insight dashboard. Every check produces anonymous demand data. The dashboard shows where travelers want to go but cannot, for example how many plans contain Petra → Wadi Rum without a car and what share of them fail, by month and origin market.

Loop 3 (operators and travelers): data stewardship. JETT, PDTRA and nature reserves update their own schedules, prices and opening hours through role-based accounts in the admin panel, and travelers confirm each leg after the trip ("Was this bus there? yes / no"). The team seeds the first 12 destinations and 10 routes; after that, maintenance shifts from the student team to the data owners and travelers, and the team moderates.

Why it fits the problem: AI parses, rules decide. The AI step only converts free text into structured days and places. A rules engine built on verified, dated data makes every

judgment about the plan, so the system cannot invent information, which is the main weakness of general assistants [6]. The AI provider sits behind an adapter interface, so the team can choose or replace it without changing the engine.

Added value: (1) it fixes the plan the traveler already has, or builds one that is feasible from the start; (2) real cost in JOD for every leg; (3) a Jordan Pass calculator built into the plan; (4) every piece of information shows its last-verified date and source; (5) it steers travelers towards less-visited sites that fit their trip.

Alignment with the National Tourism Strategy

The Ministry of Tourism and Antiquities' National Tourism Strategy 2021–2025 states that "a Jordan-wide Tourism Digital Map/App will be developed" (p. 12) and calls for a Tourism MIS/Dashboard and the digitization of MoTA services (p. 19) [18]. Darb's admin panel and its anonymous issue data form that dashboard for independent travel: which routes travelers plan, which fail and why. The strategy also uses domestic tourism "as a tool to offset seasonality" (p. 14) [18]; Darb's feasible-discovery flow supports the same goal by guiding foreign visitors to lesser-visited sites that fit their trip.

Assumptions and Validation

– **Assumption 1:** first-time travelers rely on ready-made plans from AI or blogs.

Validation: a short survey of 30+ international travelers through travel groups and hostels in Amman.

– **Assumption 2:** a large share of ready-made plans contain real transport issues.

Validation: generate 20 Jordan itineraries with ChatGPT, check them manually and record the share of days with issues. This test is part of the competitor evaluation the team has started.

– **Assumption 3:** users trust and adopt the corrections. Validation: usability testing of the prototype with 5–8 users, measuring the share who accept the corrected plan.

– **Assumption 4:** transport data can be kept up to date at low cost. Validation: a one-month field and phone verification trial on 10 key routes, measuring the time needed per update.

– **Assumption 5:** operators will maintain their own data if it brings them demand.

Validation: one meeting with JETT or PDTRA using a mock-up of the demand dashboard.

3. Project Scope and Requirements

In Scope and Out of Scope

In scope for version 1: responsive English website; plan input (paste text or guided questions); "Build a plan" feasible discovery (interests, pace, days, car / no car, with destination cards labeled by feasibility); Reality Check for the 12 top destinations (Amman, Jerash, Ajloun, Dead Sea, Madaba & Mount Nebo, Karak, Dana, Petra, Wadi Rum, Aqaba, Umm Qais, Desert Castles); transport cost calculation; Jordan Pass calculator; corrected plan with an "if you're late" alternative for every transport leg; saving and sharing as a link or offline PDF; calendar export (.ics file and Google Calendar link, Asia/Amman time zone); a "Weather & what to pack" card (seasonal averages, live forecast when the trip starts within 7 days); a basic ministry insight dashboard; data-owner accounts in the admin panel and post-trip leg confirmation.

Language: version 1 is English-first because the foreign independent traveler carries the largest planning risk; Arabic is the first item on the roadmap.

Deferred to later phases: native mobile app; live in-trip companion mode; direct booking and payment for transport; additional languages (Arabic first, then German and French); local experiences with hosts; accounts for tour organizers.

Requirement / Feature	Priority	Acceptance Criterion
Plan input by pasting text or guided questions	High	A 5-day plan is converted into correct days and places with 90% accuracy on a test sample
Reality Check with color-coded issues and fixes	High	Every issue shows a reason, fix and cost; result appears in under 8 seconds
Corrected plan and total cost calculation	High	Total of transport and tickets matches stored data with no discrepancies
Jordan Pass calculator	Medium	Selects the cheapest tier and shows savings based on stay length and sites
Save, share and offline PDF	Medium	The link opens the plan on another device; the PDF loads in under 5 seconds
Calendar export (.ics / Google Calendar)	Medium	Every day and leg appears as an event with correct Asia/Amman

		times; the file opens on iOS, Android and Outlook
Weather & packing card (seasonal averages; live forecast ≤ 7 days)	Medium	Each plan day shows expected temperatures and at least one site-specific tip; the forecast is shown when the start date is within 7 days
Build a plan: interests, pace, days, car $\rightarrow$ feasible destination cards	Medium	Every suggested destination is labeled "fits your trip" or "needs +1 day or a car"; the generated plan scores 90+
Ministry insight dashboard	Medium	Shows plan counts and failure rates per route (e.g., Petra $\rightarrow$ Wadi Rum without a car), updated daily, with no personal data

Quality Requirements and Constraints

- **Performance:** check results in under 8 seconds for a 7-day plan; pages load in under 3 seconds on 4G.
- **Usability:** a new user reaches the check result in 3 steps or fewer, with no login.
- **Compatibility:** latest two versions of Chrome, Safari and Firefox; full display on screens 360 px wide and above.
- **Data accuracy:** every price and schedule carries a last-verified date; information older than 90 days is labeled "estimated".
- **Constraints:** local transport data has no official API; the team seeds it from operator systems and field checks, then data owners maintain it through the admin panel. Limited operating budget; a team of 4 students.

4. Technical Architecture

Main Components

1. **Static web app:** plain HTML, CSS and JavaScript with no framework, containing the input, Reality Check, corrected plan and map pages. It is the simplest option to build and host, loads fastest, serves real HTML that search engines and AI assistants can read (SEO/GEO), and matches the team's skills.
2. **Node.js API (Express) on Hostinger:** a lightweight server that runs the Reality Engine server-side, applies rate limiting and saves plans; the Hostinger Business plan supports Node.js apps [9].
3. **Plan Parser (rule-based, first pass and fallback):** handles the fast path: splits pasted text on "Day N" patterns and matches place names and aliases against the destination table (fuzzy match). Text it cannot parse confidently goes to the AI Parsing Service; if the AI step also fails or is unavailable, the parser's partial result is used and the user is offered guided questions. The user always confirms the parsed days before checking.
4. **AI Parsing Service (model-agnostic):** converts free-text itineraries into structured JSON (days, places, order) through an adapter interface; only place names already in the destination table are accepted. The team will pick the provider and model at build time by cost, latency and English/Arabic quality, and any provider fits behind the same interface. The AI makes no decisions; the Reality Engine makes every judgment.
5. **Reality Engine:** code-based rules that check each day: availability of transport at the required time, total travel plus visit hours, site opening hours and cost. It proposes alternatives ranked by price and time, and explains each issue with template sentences generated by the rules (AI wording may be added later as an option).
6. **Database:** PostgreSQL on Supabase, holding places, routes, schedules, prices, Jordan Pass rules and saved plans.
7. **External services:** Google Maps Platform for maps, distances and travel times [10], and an LLM API (provider to be selected) behind the AI parsing adapter.
8. **Admin panel (role-based):** the team seeds and moderates data; data owners (JETT, PDTRA, reserves) update their own records with a last-verified date; a ministry view shows the aggregate demand and issues dashboard.

Architecture Diagram

User (browser) ⇌ static web app (HTML, CSS, JavaScript) ⇌ Node.js API (Express) on Hostinger

Node.js API → Plan Parser (rule-based first pass: "Day N" patterns and fuzzy place matching against the destination table)

Plan Parser → AI Parsing Service (provider to be selected): text → JSON; falls back to the rule-based result and guided questions

Node.js API → Reality Engine ⇌ Supabase database (places, routes, prices, rules)

Reality Engine → Google Maps Platform (distances and travel times when needed)

Admin panel → database (updates by the team and data owners; read-only demand dashboard for the ministry)

Result → Node.js API → web app (Reality Score, issues, fixes, corrected plan)

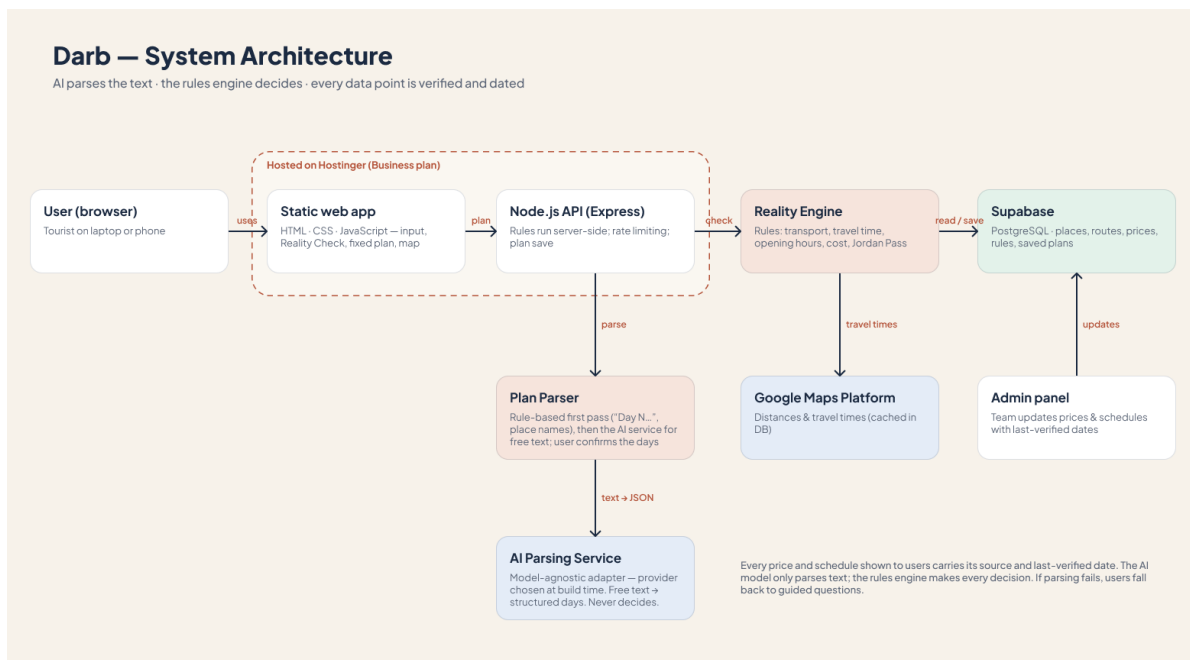


Figure 1 — System architecture (also on the "Architecture" page of the Figma file).

Key Process Flow

Process: checking a pasted plan

1. The user pastes a 5-day plan, indicates no car, and clicks "Check my plan".
2. The web app sends the text to the Node.js API. The Plan Parser tries the "Day N" fast path; because this plan is free text, it is passed to the AI Parsing Service, which returns structured JSON matched against the destination table: Day 1 Amman, Day 2 Petra, Day 3 Wadi Rum and Aqaba, and so on. The user confirms the parsed days (or fixes them) before anything is checked.
3. The Reality Engine matches each place to its database record and retrieves transport options between each pair of consecutive places.

4. The engine applies the rules. The Amman → Petra leg is matched to the only JETT departure (Abdali 06:30 → Petra 10:30, 4 h, 10 JOD [14]), so the engine marks a plan that starts sightseeing in Petra at 9:00 that day as "red". Getting from Petra to Wadi Rum without a car requires a taxi or pre-arranged transfer, because JETT offers no bookable Wadi Rum service [4][14]; the engine marks this leg "yellow" and adds its cost.
5. The engine calculates the total trip cost and compares site tickets with Jordan Pass prices [7][8].
6. The rules engine phrases each issue as a template sentence (e.g., "No bookable bus reaches Wadi Rum from Petra; use a taxi or pre-arranged transfer", with its estimated cost). Darb saves the plan, the result and an anonymous issue record (route, severity, car / no car); the issue record feeds the ministry dashboard.
7. The web app shows the Reality Score, issues and fixes; the user clicks "Fix all" to get the corrected plan and saves it as a link or PDF.

5. Technologies and Tools

Area	Technology / Tool	Reason for Choice
User interface	HTML + CSS + JavaScript (static, no framework)	Simplest to build and host; fastest load; real HTML for SEO/GEO; matches the team's skills
Processing and logic	Node.js (Express) API + JavaScript rules engine + rule-based Plan Parser + AI parsing adapter (provider TBD)	AI parses, rules decide: consistent, verifiable results; the AI provider sits behind an adapter and is chosen at build time by cost, latency and English/Arabic quality
Data storage	Supabase (PostgreSQL)	Relational database suited to places and routes, with built-in auth, access policies and a free starter plan [11]
Design and collaboration	Figma + GitHub + Notion	Interactive design, code management and review, data and task documentation
Hosting	Hostinger (Business plan) + Google Maps Platform	Hosts the static site and Node.js API [9]; maps usage stays within the free monthly cap with caching [10]
Testing and version control	Playwright + Vitest + GitHub Actions	Automated tests for the core flow and rules before every deployment

Comparison of Alternatives

Key decision: AI parses, rules decide. An AI step converts free text into structured days and places; deterministic rules on verified data make every judgment. The language model is never asked to generate, judge or correct the plan.

- **Fit:** a language model alone can invent schedules and places [6]; here its output is limited to places already in the destination table and is confirmed by the user, while rules give consistent results with no hallucinations, and every result can be traced to a dated record.

- **Cost:** the rules engine and database lookups have no per-request fee; the only per-request cost is a small AI parsing call, plus a fixed hosting plan and maps usage within the free monthly cap [9][10]. The parsing budget is set out in the financial model.

- **Team expertise:** the team is experienced with JavaScript (browser and Node.js) and PostgreSQL; writing rules and calling an LLM API through an adapter requires no specialized machine-learning skills.

- **Maintainability:** changing a price or schedule means updating a database record, with no retraining or prompt changes, because the AI only parses text. The parsing provider or model can be swapped behind the adapter without touching the engine.

Frontend alternative: we chose a static HTML, CSS and JavaScript app over a JavaScript framework: it is the simplest to build and host, loads fastest, serves real HTML for search engines and AI answers (SEO/GEO), and matches the team's skills.

Maps alternative: we compared Google Maps Platform with OpenStreetMap + OSRM. We chose Google initially because the monthly free usage (10,000 requests per Essentials SKU [10]) covers version 1 and integration is faster. OSRM remains a free fallback if usage grows.

Dependencies and Licenses

- Node.js and Express: open-source, MIT license. The frontend is plain HTML, CSS and JavaScript with no framework.

- Hostinger Business plan: hosts the static site and the Node.js API (Node.js apps are supported from the Business plan); USD 3.79/month introductory, USD 16.99/month on renewal [9]. API keys are stored only as server-side environment variables.

- Supabase: free plan during development; Pro plan at USD 25/month at launch [11].

- Google Maps Platform: free monthly cap of 10,000 requests per Essentials SKU (5,000 Pro, 1,000 Enterprise). Beyond the cap, prices per 1,000 requests (0–100K tier) are USD 7.00 for Dynamic Maps, 2.00 for Static Maps, 5.00 for Compute Routes Essentials, 5.00 for Place Details Essentials, 5.00 for Geocoding and 2.83 for Autocomplete [10]; we must follow its display and attribution terms.

- LLM API (provider to be selected): pay-as-you-go; budgeted below. Called only through the parsing adapter, under the chosen provider's terms of use.

- iCalendar (RFC 5545): open standard; no license or fee. The .ics files are generated by Darb's own code.

- Weather API: free tier; provider to be confirmed at build time. Open-Meteo is the baseline: a free forecast API with no API key, under its terms of use for non-commercial use [21].

- Site and price data: from cited official and published sources plus the team's field verification. No protected content is copied from other websites.

6. Data and Integrations

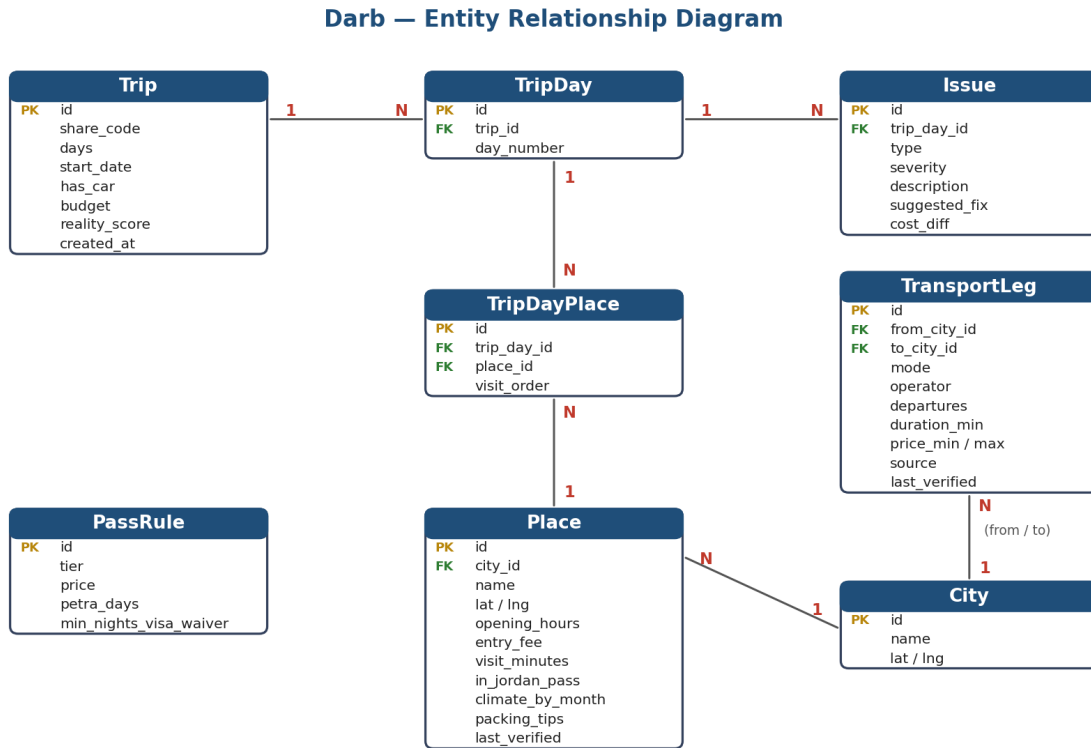
Data / Entity	Source	Use
Tourist sites (opening hours, prices, visit duration)	Official sources [7][8], data owners (PDTRA, reserves) via the admin panel + field verification	Check visit feasibility and calculate cost
Transport routes, schedules and prices	Operator booking systems (e.g., JETT [14]) and operators via the admin panel; guides [3][4][5][19] only as flagged estimates	Check transport availability at the required time and suggest alternatives
Jordan Pass rules and prices	Official site and published guides [7]	Calculate savings and choose the best tier
Seasonal climate averages and packing tips per destination	Published climate normals seeded by the team; live forecast API when the trip starts within 7 days	Packing tips and seasonal timing rules
User plans and post-trip leg confirmations	User input	Parsing, checking, saving, sharing and confirming data accuracy
Anonymous issue records (route, severity, car / no car)	Generated by the Reality Engine	Ministry demand and issues dashboard

Data Structure and Relationships

- **City:** id, name, coordinates.
- **Place:** id, city_id, name, coordinates, seasonal opening hours, entry fee, average visit duration, included in Jordan Pass (yes/no), climate_by_month (average day/night °C for each month), packing_tips, last verified date.
- **TransportLeg:** id, from_city_id, to_city_id, mode (JETT, local bus, taxi, rental car, private driver), operator, departure times, duration, min/max price, source, last verified date.
- **Trip:** id, share code, number of days, start date, has car (yes/no), budget, Reality Score, created date.
- **TripDay:** id, trip_id, day number.
- **TripDayPlace:** id, trip_day_id, place_id, visit order (links days to the places visited, in order).
- **Issue:** id, trip_day_id, type (transport, time, closure, cost), severity (red, yellow, green), description, suggested fix, cost difference.

– **PassRule:** tier, price, Petra days included, minimum nights for the visa waiver (reference table).

Relationships: a Trip has many TripDays; a TripDay has many Issues and many TripDayPlaces; each TripDayPlace points to one Place; each Place belongs to one City; each TransportLeg connects two Cities (from and to), and the Reality Engine uses it to link the places of consecutive days.



PassRule is a reference table used by the Reality Engine to calculate Jordan Pass savings from Place.in_jordan_pass and trip length.

Figure 2 — Entity Relationship Diagram (ERD) of the Darb database.

Integration with Systems and Services

– **Google Maps Platform:** receives the coordinates of two places and returns distance and travel time. Results are cached in the database to reduce requests; if the service fails, stored travel times are used. At about 2,000 plans per month with cached travel times, usage stays under the free cap; in the worst case with no caching (about 10 route calls per plan = 20,000 calls), the cost would be about USD 50/month ($10,000 \times 5.00 / 1,000$) [10], so caching is mandatory.

– **LLM API (AI parsing, provider to be selected):** the Node.js API sends the pasted itinerary text through an adapter interface and receives structured JSON (days, places, order), validated against the destination table. The API key is stored server-side only. If the call fails or returns invalid JSON, it is retried once; if it fails again, Darb falls back to

the rule-based parser and guided questions. The provider and model are not fixed at design time and can be changed without changing the Reality Engine.

- **Weather API (provider to be confirmed at build time; Open-Meteo is the baseline because it needs no key and has no cost [21]):** used only when a trip starts within 7 days, to replace seasonal averages with a live 7-day forecast. Forecasts are cached once a day per destination (about 12 calls per day), so traffic does not grow with the number of plans. If the call fails, Darb falls back to the seasonal averages stored in the destination table, and the card says so.

- **Calendar export (iCalendar, RFC 5545):** the Node.js API generates the .ics file from the saved plan, with no external service and no cost: one event per day and one per transport leg, with departure time, duration, place address and coordinates, cost and the "if you're late" alternative in the event notes, a reminder 45 minutes before each departure, and the Asia/Amman time zone. The file works with Apple Calendar, Outlook and any phone; a "Google Calendar" link pre-fills the same events, and the offline PDF includes the calendar link.

- **Supabase:** database and storage with daily backups; if the connection fails, a clear message is shown and the draft plan is kept in the browser.

7. UX Design and Interfaces

User Journey

Scenario: Sarah, 29, from Germany, first visit to Jordan, 5 days, no rental car. She has a plan from ChatGPT.

1. She opens Darb and clicks "Paste your plan".
2. She pastes the plan and selects Queen Alia Airport, October, no car.
3. The Reality Check screen appears with a low score: Day 3 is red because getting to Wadi Rum at the planned time is not possible without a car; Day 4 is yellow because it combines Jerash and the Dead Sea.
4. She opens each issue to see the reason and the fix: a pre-arranged transfer with an estimated price, or reordering the days.
5. She sees the Jordan Pass card: she saves about 33 JOD compared with buying the visa and tickets separately [7].
6. She clicks "Fix all" and gets a corrected plan with every transport leg (e.g., JETT Abdali 06:30 → Petra 10:30, 10 JOD [14]) and the total trip cost; the weather card reminds her that Wadi Rum nights are cold even in October, so she packs a warm jacket.
7. She saves the plan as a link, downloads a PDF that works offline, and adds the trip to her phone calendar, with a reminder 45 minutes before each departure.

Scenario 2 (Build a plan): Ahmad, 27, from the Netherlands, 6 days, no car and no plan at all.

1. He clicks "Build a plan" and picks interests (history, nature / hiking, desert, food, relaxation), a relaxed pace, 6 days and no car.
2. Darb shows destination cards, each labeled "fits your trip" or "needs +1 day or a car", and puts less-visited feasible sites such as Umm Qais, Ajloun, Dana and As-Salt next to Petra and Wadi Rum.
3. He selects his places; the engine orders them geographically and links verified transport for every leg.
4. The plan starts at a Reality Score of 90+, and he continues to the same Fixed Plan, cost and Save & Share screens as Sarah.

Main Screens

1. **Landing:** explains the value in one sentence, with two buttons: "Paste your plan" and "Build a plan".

2. **Input:** a box to paste the plan, or guided questions (days, airport, car, month, budget).
3. **Reality Check:** the key screen, with the Reality Score, a color-coded day timeline, details of each issue and its fix, the Jordan Pass card and a route map. Main action: "Fix all".
4. **Fixed Plan:** the final day-by-day program with every transport leg (mode, time, price), a full cost breakdown, a "Weather & what to pack" card (expected day/night temperatures and site-specific tips) and an "Add to calendar" button.
5. **Leg Detail:** comparison of transport options between two cities by time, price and notes.
6. **Save & Share:** share link, PDF download, offline access and a calendar row (Google, Apple or .ics) that adds one event per day and per transport leg.
7. **Your interests (Build a plan):** interests, pace, days and car / no car, then destination cards labeled "fits your trip" or "needs +1 day or a car".
8. **Ministry dashboard:** plan volumes by route, failure rates (e.g., Petra → Wadi Rum without a car), most-requested but infeasible sites, and data freshness per data owner.

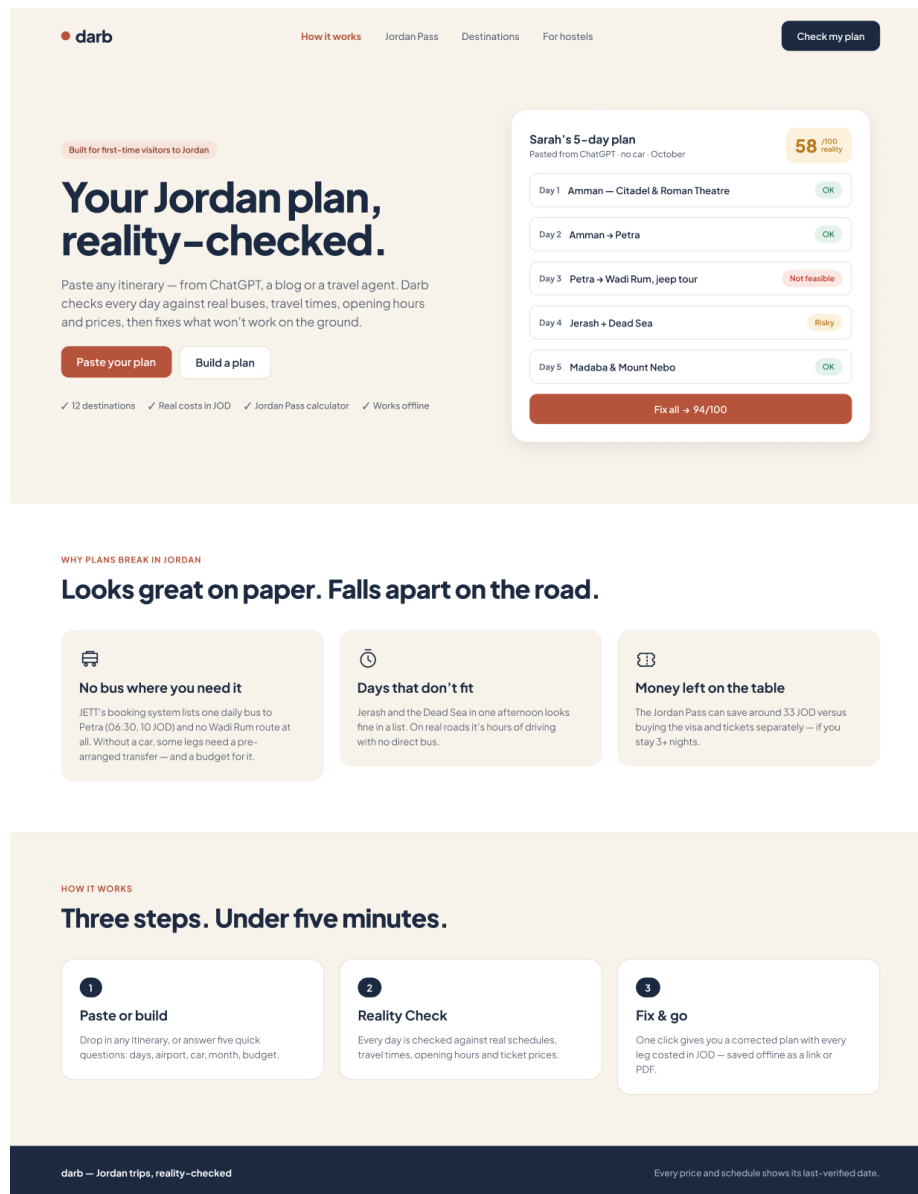


Figure 3 — Landing.

darb

How it worksJordan PassDestinationsFor hostels

Check my plan

1 Your plan

2 Reality Check

3 Fixed plan

Tell us about your trip

Paste your planBuild with questions

Day 1 – Arrive in Amman. Visit the Citadel and the Roman Theatre.
Day 2 – Drive or take a bus to Petra. Explore the Siq and the Treasury.
Day 3 – Morning at Petra, then head to Wadi Rum for a sunset jeep tour and desert camp.
Day 4 – Visit Jerash in the morning and float in the Dead Sea in the afternoon.
Day 5 – Madaba mosaics and Mount Nebo, then fly home.
Pasted from ChatGPT - 412 characters

Arrival airport
Queen Alia (AMM)

Month
October

Travelers
1 adult

Budget
Mid-range

Will you have a car?

Yes, renting a car

✓ No car – public transport & taxis

Check my plan →

No login needed. Your plan is saved privately under a random link.

What we check for every day

✓ Transport
Is there a real bus, transfer or taxi at the time you need it?

✓ Travel time
Real road times between places – not straight-line guesses.

✓ Opening hours
Seasonal hours and closures at each site.

✓ Costs
Tickets and transport in JOD, with the last-verified date.

✓ Jordan Pass
Whether the Pass saves you money for this plan.

Figure 4 — Input: pasted ChatGPT plan, trip details and the no-car option.

21

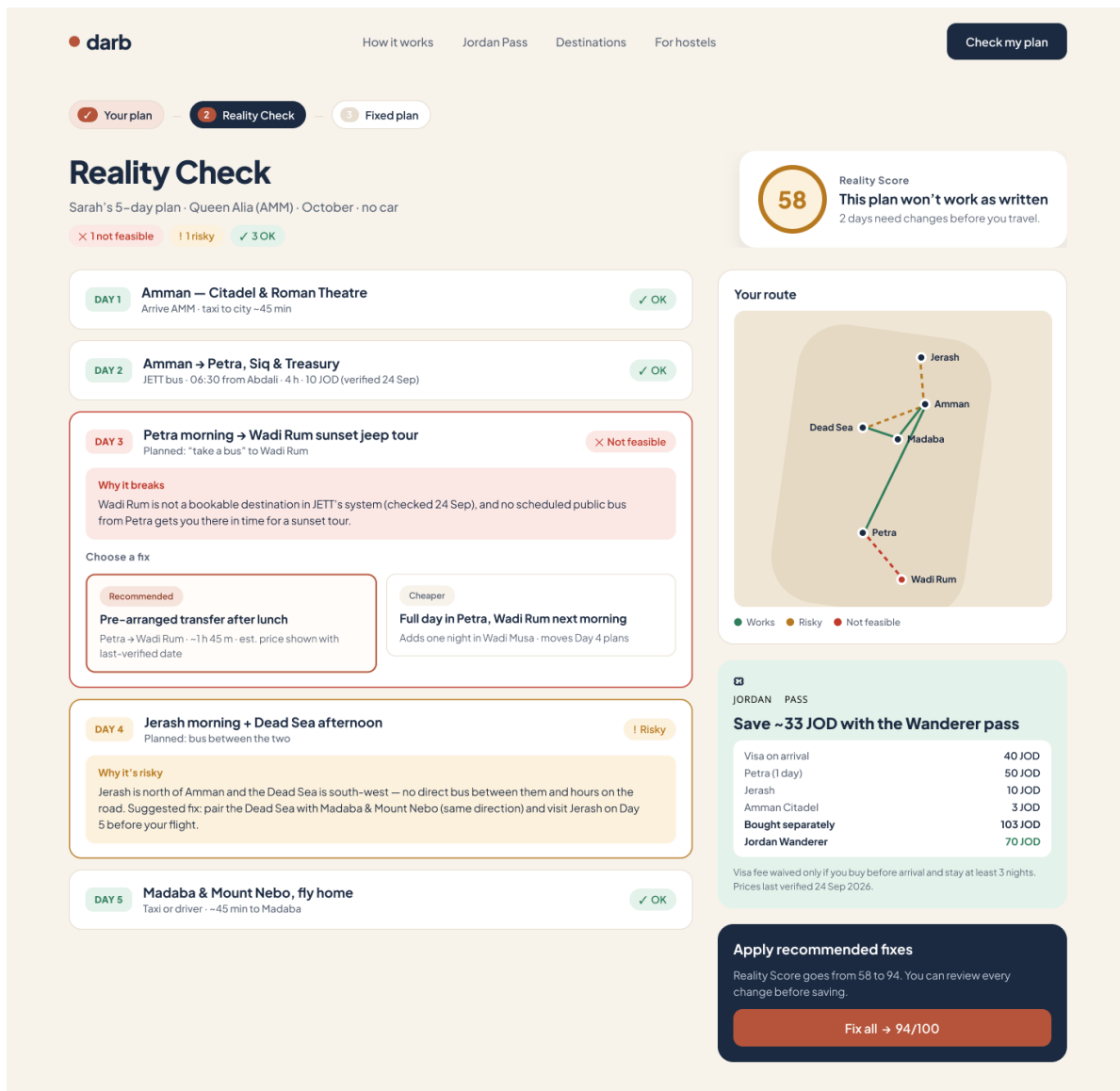


Figure 5 — Reality Check: score 58/100, one infeasible day, one risky day, route map and Jordan Pass savings.

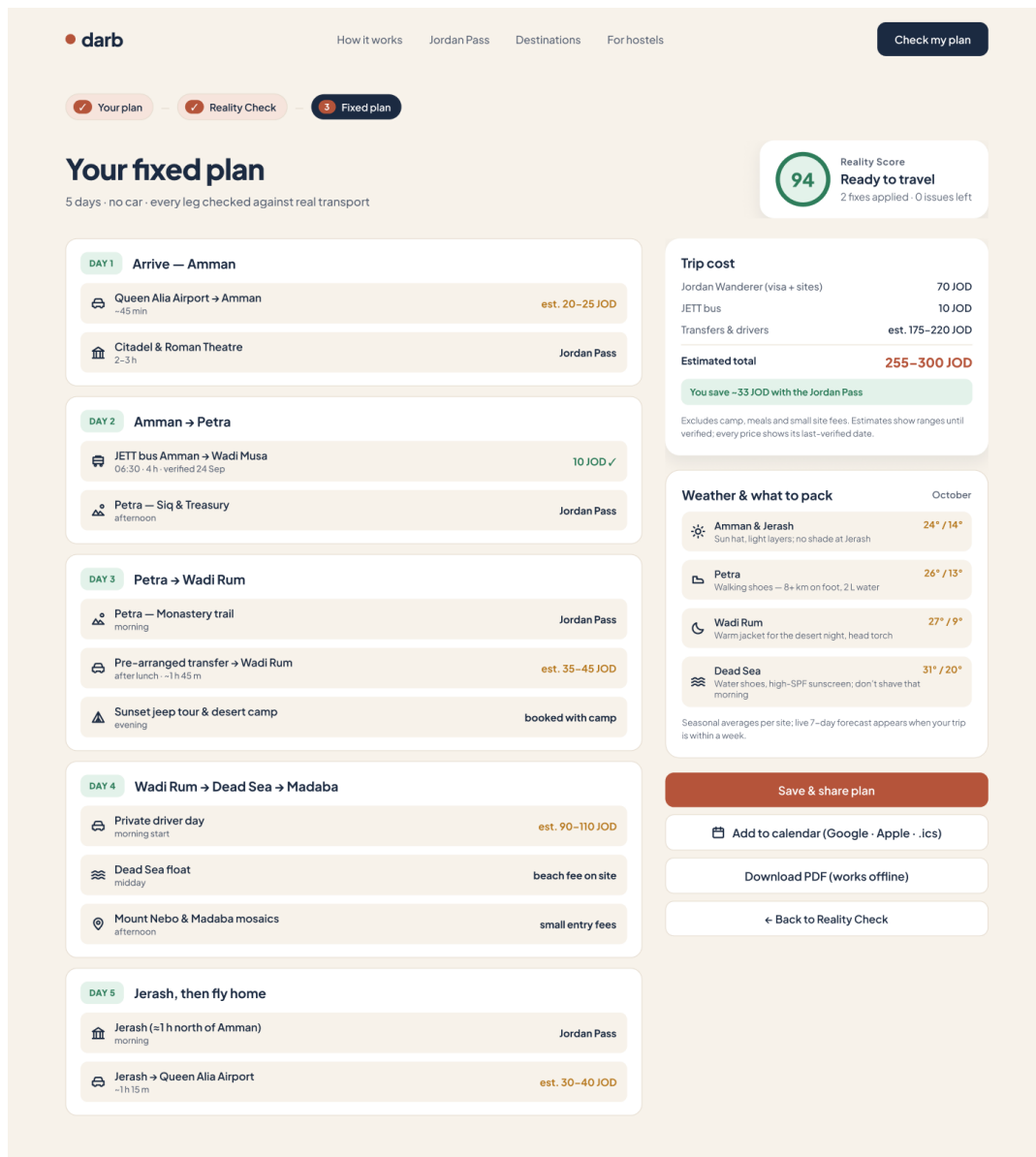


Figure 6 — Fixed plan: score 94/100 with every leg and cost (JETT leg shows the verified 06:30 · 4 h · 10 JOD [14]); estimates shown as ranges until verified; the "Weather & what to pack" card and the "Add to calendar" button.

[How it works](#)
[Jordan Pass](#)
[Destinations](#)
[For hostels](#)

Check my plan

← Back to Day 3

DAY 3 · TRANSPORT LEG

Petra (Wadi Musa) → Wadi Rum

You need to arrive before ~4 pm for a sunset jeep tour. You don't have a car.

×

No JETT service to Wadi Rum — the public bus option your plan assumed isn't available at this time.

OPTION	TIME	COST	ARRIVES	NOTES
Pre-arranged transfer Recommended	~1 h 45 m	est. 35–45 JOD per car · last verified 24 Sep	✓ In time for sunset	Book through your camp or a licensed driver; fixed price agreed in advance.
Taxi from Wadi Musa	~1 h 45 m	Negotiated on the day	✓ Usually in time	Agree the fare before leaving; availability varies in the afternoon.
Bus via Aqaba + taxi	Much longer	Lower fare, extra taxi	× Misses sunset	Adds a change in Aqaba; not practical for this day.
Rental car	~1 h 30 m	25–30 JOD / day + fuel · last verified 24 Sep	✓ In time	Not in your plan (no car). Drivers must be 25+.

Use recommended option

Compare on map

Prices marked "est." are ranges until verified by the Darb data team. Every option shows its source and last-verified date.

Figure 7 — Leg detail: comparing options for Petra → Wadi Rum without a car.

[How it works](#)
[Jordan Pass](#)
[Destinations](#)
[For hostels](#)

Check my plan

✓ Your plan

✓ Reality Check

Your fixed plan

5 days · no car · every leg checked against reality

DAY 1

Arrive — Amman
Queen Alia Airport → Amman
~45 min
Citadel & Roman Theatre
2–3 h

DAY 2

Amman → Petra
JETT bus Amman → Wadi Musa
06:30–4 h · verified 24 Sep
Petra — Siq & Treasury
afternoon

DAY 3

Petra → Wadi Rum
Petra — Monastery trail
morning
Pre-arranged transfer → Wadi Rum
after lunch · ~1 h 45 m
est. 35–45 JOD

Reality Score

94

Ready to travel

2 fixes applied · 0 issues left

70 JOD

10 JOD

est. 175–220 JOD

255–300 JOD

JORD with the Jordan Pass

meals and small site fees. Estimates show ranges until ce shows its last-verified date.

what to pack

October

& Jerash

24° / 14°

Walking shoes — 8+ km on foot, 2 L water

26° / 13°

Wadi Rum

Warm jacket for the desert night, head torch

27° / 9°

Dead Sea

Water shoes, high-SPF sunscreen; don't shave that morning

31° / 20°

Save & share your plan

×

Sarah's 5-day Jordan plan · Reality Score 94 · no login needed

darb.app/t/k7Q9-mX2p-4Rz8

Copy link

Only people with this link can view the plan.

Download PDF

Day-by-day plan with every leg and cost

Save offline

Open the plan with no signal in Wadi Rum

Email to me

Get the link and PDF in your inbox

Add the whole trip to your calendar

One event per day and per transport leg, with departure times, addresses and the "if you're alternative in the notes."

Google

Apple

.ics

Trip Pass · 3.5 JOD one-time

Unlocks offline PDF and price updates before you travel.

Unlock

Done

24

Figure 8 — Save & share: private link, offline PDF, calendar export (Google, Apple, .ics) and the Trip Pass.

darb

How it works

Jordan Pass

Destinations

For hostels

Check my plan

Build a plan that actually works

No itinerary yet? Tell us what you like — we only suggest places you can really reach.

1 · Your interests

2 · Your days

3 · Your plan

What do you enjoy?

History & ruins ✓

Nature & hiking ✓

Desert & stars ✓

Food & culture

Beach & relaxation

Faith & pilgrimage

Days

Arrival

Month

Car

5

Queen Alia (AMM)

October

No car

Car

Pace

Relaxed

Balanced ✓

Packed

Places that fit your trip

Petra

Jordan Pass covers entry · JETT bus 06:30 from Amman

Fits your trip

Added ✓

Wadi Rum

Needs a pre-arranged transfer from Petra · est. 35 JOD

Needs +1 day or a car

Added ✓

Jerash

Roman city · JETT daily tour from Amman

Fits your trip

Added ✓

Umm Qais

Hidden gem

Lesser-visited · Gadara ruins · 1h from Irbid

Fits your trip

Added ✓

Dana Biosphere Reserve

Hidden gem

Canyon hikes & eco-lodge · needs +1 day

Needs +1 day or a car

+ Add

As-Salt

Hidden gem

UNESCO town · half day from Amman by minibus

Fits your trip

+ Add

Your plan so far

Day 1 Arrive Amman · Citadel & downtown

Day 2 Jerash & As-Salt

Day 3 Petra (full day)

Day 4 Wadi Rum · camp overnight

Day 5 Umm Qais — 2 h 30 from Petra area

▲ Umm Qais moves to Day 2 (next to Jerash) — saves ~5 h on the road

Reality Score (draft)

91

Build my plan →

Why these?

We rank places by fit to your interests AND by verified transport from your previous stop. Lesser-visited sites get a boost when they fit.

Figure 9 — Build a plan: interests and constraints first, then only destinations that fit; lesser-visited sites are surfaced when feasible.

25

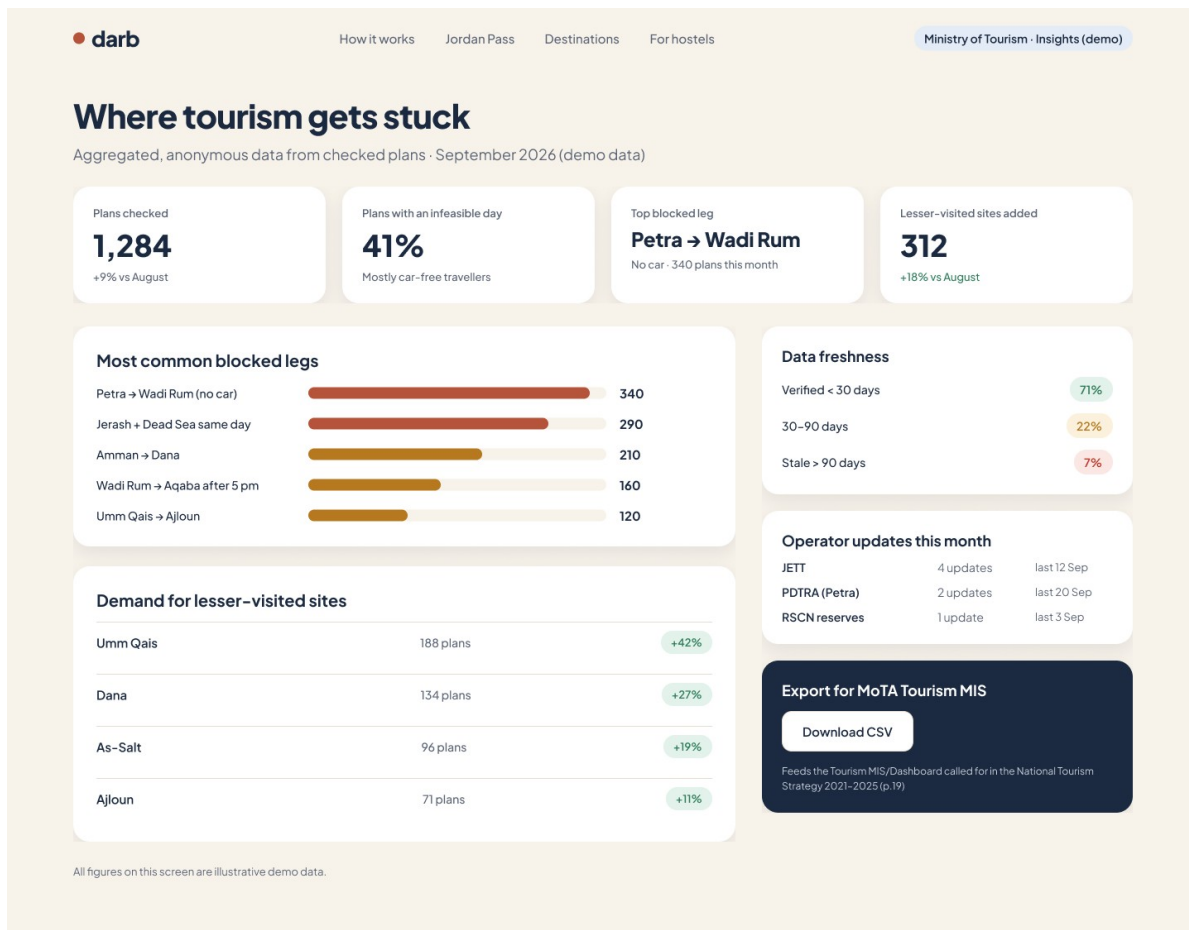


Figure 10 — Ministry dashboard (demo data): blocked legs, demand for lesser-visited sites, data freshness and export to the Tourism MIS.

Figma Design Link

Figma file (design + clickable prototype):

<https://www.figma.com/design/QFKG3YDunp7qlfcwvRswlt>

Usability and Accessibility

- **Navigation:** a linear 3-step flow (input → check → plan) with a clear back button at every step and no mandatory login.
- **Text:** plain English; each issue is explained in one sentence alongside its fix.
- **Color:** color is never the only signal. Every status has an icon and a label (Not feasible, Risky, OK), and contrast is at least 4.5:1 per WCAG AA.
- **Screen sizes:** mobile-first responsive design; the map becomes a separate tab on small screens.

- **Diverse needs:** keyboard navigation, alt text for images, and text enlargement without breaking the layout.

8. Security, Privacy and Testing

Security and Privacy

- No login is required for the core service. Plans are saved under a long random code, and only people with the link can view them.
- No sensitive personal data is collected; a clear privacy policy is shown, and all data travels over HTTPS.
- Server keys (Google Maps routing, Supabase service key, LLM API key) are stored only as environment variables on the Node.js API and never reach the browser; the browser map key is restricted to Darb's domain. Only the itinerary text is sent for AI parsing, with no personal data.
- Database access is controlled with Row Level Security policies. The admin panel is limited to team accounts with two-factor authentication.
- Per-user rate limiting prevents abuse and unexpected usage costs; the database is backed up daily.

Test Case	Expected Result	Success Criterion
Paste a 5-day plan with an impossible no-car transfer	Day flagged red with an alternative and its cost	Issue detected in 100% of known test cases
Paste empty or non-travel text	The parser returns a clear message asking for a valid plan or offering guided questions	No technical error and no fake result
50 simultaneous check requests	All results returned without failure	Average response time under 8 seconds
Access the admin panel or another user's plan without permission	Access denied	No data leakage; RLS policies enforced
Route data older than 90 days	Shown to the user as "estimated" and flagged in the admin panel	100% of stale rows flagged
Export a 5-day plan to .ics	One event per day and per transport leg, with correct times (Asia/Amman)	Opens without errors in two calendar apps
Trip starting in 3 days	Live forecast replaces the seasonal averages	Forecast date matches the trip date; falls back to averages if the API fails

User Testing and Feedback Handling

- **Testers:** 5-8 international travelers from hostels in Amman and online travel groups, plus 3 tourism workers (a tour guide, a receptionist and a driver).
- **Collecting feedback:** test sessions with defined tasks (check your plan, fix Day 3), recording time and errors, a short survey afterwards and an in-app feedback button.
- **Prioritizing fixes:** the team classifies each piece of feedback by impact (blocks the task, slows it, cosmetic) and frequency; blockers come first.
- **Re-verification:** after each batch of fixes, the team repeats the same tasks with new users and compares time and success rate.

9. Implementation and Launch Plan

Phase	Duration	Owner	Deliverable
Analysis	2 weeks	Team Lead	Requirements document and initial database for 12 destinations and 10 routes
Design	2 weeks	UI/UX Designer	Interactive Figma design and first usability test results
Development	5 weeks	Two developers	Working version of the full flow (input, check, corrected plan, save)
Testing	2 weeks	Developer / QA	Test report, user testing and a closed fix list
Launch	1 week	Team Lead	Live site on a domain, error monitoring and a data update plan

Dependencies and Delivery Risks

- **Dependencies:** design depends on the analysis results; Reality Engine development depends on data being ready for the 12 destinations; testing depends on a complete version of the core flow.
- **Risk 1 (transport data is hard to collect):** start with the 10 most important routes, document the source and verification date, and add the rest gradually.
- **Risk 2 (prices and schedules change):** set an expiry date for every data point, flag data older than 90 days as "estimated", and move maintenance to data owners and traveler confirmations.
- **Risk 3 (academic workload on the team):** split tasks weekly with a weekly review, and define the minimum feature set for each phase.
- **Risk 4 (parsing accuracy, rule-based + AI):** pasted plans vary in format. Mitigation: the user confirms the parsed days before checking; guided questions are always available; the AI provider can be swapped without changing the engine.

Launch and Maintenance

- **Environment:** Hostinger (Business plan) for the static site and the Node.js API, Supabase for the database, with separate development, staging and production environments.
- **Deployment:** every change goes through code review on GitHub and automated tests (GitHub Actions), then deploys automatically to a staging subdomain on Hostinger, and to production after approval.
- **Monitoring:** error logging, alerts when external service calls fail, and weekly tracking of maps usage against the free cap and of AI parsing spend against its budget.
- **Support and updates:** The team lead owns operations and technical support. After the seeding phase, data owners update their own records; the team moderates changes, reviews traveler confirmations and runs a monthly report of data older than 90 days.
- **Competition phases:** later phases require a working web build, so the team keeps the code on GitHub with automated tests from the first week.

10. Implementation Cost

Currency: Jordanian Dinar (JOD) Implementation period: 12 weeks Estimate date: 24/09/2026

Item	Quantity	Unit Price	Total
Requirements analysis and tourist research	30 hours	5 JOD	150
UX and UI design in Figma	50 hours	5 JOD	250
Frontend development	150 hours	5 JOD	750
Backend, Reality Engine and database	200 hours	5 JOD	1,000
Transport and site data collection (desk research)	60 hours	5 JOD	300
Field verification of key routes	10 trips	15 JOD	150
Testing and user testing	50 hours	5 JOD	250
Deployment and production setup	10 hours	5 JOD	50
Domain (1 year)	1	15 JOD	15
Base implementation cost	—	—	2,915
Contingency reserve	10% of base	—	291.5
Total implementation budget	—	—	3,206.5

Basis of Estimate and In-Kind Contributions

Calculation: line total = quantity × unit cost. Contingency = 2,915 × 10% = 291.5. Total implementation budget = 2,915 + 291.5 = **3,206.5 JOD**. The reserve covers limited unforeseen work; actual use will be recorded.

- **Hourly rate:** 5 JOD, based on the median software developer pay in Amman (about 717 JOD/month [12]) divided by roughly 160 hours and rounded. Total effort is 550 hours over 12 weeks (about 11–12 hours per member per week).
- **In-kind contributions:** all work hours (2,750 JOD) are contributed unpaid by the four team members and are recorded as an in-kind contribution, which reduces the cash requirement. Cash items are field verification (150 JOD) and the domain (15 JOD) = 165 JOD, plus the contingency reserve.
- **Hosting and services:** published provider prices [9][10][11], converted at the fixed rate of USD 1 = 0.709 JOD.
- **Calendar and weather:** calendar export and weather tips add no recurring cost (iCalendar is generated locally; the weather API free tier covers ~12 destinations × daily refresh).
- **Taxes and fees:** figures exclude any applicable taxes or fees; these will be verified and added before approving a real budget.

11. Operating Budget and Funding

Budget period: one month after launch Currency: Jordanian Dinar (JOD) Expected case: 120 paid trips per month

Operating Item	Cost per Period	Basis of Calculation
Hosting and database	30 JOD	Hostinger Business USD 16.99 renewal price (12.0 JOD) [9] + Supabase Pro USD 25 (17.7 JOD) [11]
Maps API	0 JOD	Within the Google Maps free monthly cap with cached travel times [10]
Data verification and maintenance	100 JOD	20 hours × 5 JOD (price/schedule updates and fixes)
Customer support	20 JOD	4 hours × 5 JOD
Marketing and outreach	100 JOD	Content for search and AI answers (SEO/GEO) plus a small social budget targeting travelers planning Jordan
Tools, licenses and domain	5 JOD	Monthly allocation of domain and tool renewals
Fixed operating costs	255 JOD	Sum of the items above
Usage-related costs	36 JOD	120 paid trips × 0.30 JOD (payment fee + AI parsing calls [20])
Expected operating costs	291 JOD	255 + 36

Funding Sources

Source	JOD	Status and timing	Conditions
Team cash contribution	500	Confirmed; available before implementation	Equity contribution from the four members; no repayment
In-kind team work	2,750	Provided during implementation	550 hours valued at 5 JOD/hour [12]

PixelSite 2.0 prize	Not counted	Unconfirmed; only if the team wins	Competition rules
Incubator / grant programs	Not counted	Unconfirmed; applications after v1 launch	Provider conditions
Funding counted as available	3,250	500 cash + 2,750 in-kind	Excludes the prize and grants

The prize and grant are hypothetical possibilities, not actual awards, and no spending will be committed against them before confirmation.

Cash Requirement

Funding needed for implementation and six months of operation (expected case, excluding future sales receipts):

Budget element	Calculation	JOD
Implementation budget	Base cost + contingency	3,206.5
Six months of operation	291×6	1,746
Total funding requirement	$3,206.5 + 1,746$	4,952.5
In-kind team work	$550 \text{ hours} \times 5$	2,750
Team cash contribution	Confirmed	500
Funding gap before sales receipts	$4,952.5 - 2,750 - 500$	1,702.5

Cash requirement = implementation cost + operating costs for the period – available funding. The implementation cash items plus the full contingency ($165 + 291.5 = 456.5$ JOD) are covered by the team's 500 JOD, so the gap appears only during operations. Postponing paid marketing (100 JOD/month) for the first 6 months would reduce the gap to 1,102.5 JOD. Actual sales receipts will reduce the gap depending on their timing.

12. Financial Sustainability and Feasibility

Revenue / Sustainability Model

Darb has two sustainability paths; the choice depends on whether a public body adopts it.

Path A (public adoption): free for travelers. Revenue comes from an annual B2G license and maintenance contract with MoTA / JTB, to be priced against the operating budget of 291 JOD/month plus maintenance hours, and from B2B licenses for hostels. Operators maintain their own data, which keeps the contract small.

Path B (independent launch, modeled below):

- **Trip Pass (main revenue):** the basic Reality Check is free; the full corrected plan with offline download and pre-trip updates is a one-time payment of 3.5 JOD ($\approx$ USD 5) per trip.

- **Variable cost per paid trip:** about **0.30 JOD** = payment processing $\approx$ 0.15 JOD (an assumed 4% fee on 3.5 JOD $\approx$ 0.14, rounded up) + AI parsing $\approx$ 0.15 JOD (about 10 checked plans per paid trip at published small-model rates, $\approx$ USD 0.02 per plan [20]). The provider is not yet selected; this is a budget ceiling, not a quote.

- **Upside not included in the scenarios:** partner commissions on transport and tour bookings ($\approx$ 4 JOD per booking) and a B2B license for hostels and small hotels (25 JOD/month).

Why this model: travelers pay once, and the savings shown exceed the price. Partner revenue can be added later, but the model does not depend on it.

Formulas (all monthly JOD): Revenue = paid trips $\times$ 3.5. Operating costs = 255 + (paid trips $\times$ 0.30). Net = revenue – operating costs. The scenarios below model Path B.

How tourists find Darb (SEO and GEO)

A generic search for "Jordan" is dominated by unrelated brands (jordan.com belongs to Nike's Jordan Brand), so Darb does not compete on the country name. Travelers who are planning search for specific questions: "Amman to Petra bus", "Jordan 5 day itinerary no car", "is the Jordan Pass worth it". Each verified fact in Darb's database becomes a public, dated answer page for exactly those queries (e.g., "Amman $\rightarrow$ Petra by bus: 06:30, 4 h, 10 JOD, verified 24 Sep 2026"), hosted on a Jordanian domain (.jo) to signal geographic relevance.

The same pages are structured for AI assistants (GEO): question-answer format, schema.org markup, last-verified dates and named sources. When a traveler asks ChatGPT or another AI assistant how to get from Petra to Wadi Rum, the assistant can

cite Darb's verified data instead of an outdated blog. Data maintenance therefore doubles as the acquisition channel at near-zero marginal cost. Every answer page ends with the prompt "Check your whole plan".

Scenario	Revenue / Funding	Expenses	Net for Period
Conservative (40 paid trips)	140	267	−127
Expected (120 paid trips)	420	291	+129
Optimistic (300 paid trips)	1,050	345	+705

Feasibility Criterion and Financial Risks

Monthly operating break-even: contribution per paid trip = $3.5 - 0.30 = 3.20$ JOD.
Break-even = fixed monthly costs $\div$ contribution = $255 \div 3.20 = 79.7$, rounded up to **80 paid trips per month**.

At 80 paid trips	Calculation	JOD / month
Revenue	80×3.5	280
Costs	$255 + (80 \times 0.30)$	279
Operating surplus	$280 - 279$	1

This threshold covers monthly operations only; it does not recover the implementation cost. For context, Jordan received about 4 million visitors in the first 7 months of 2025, including about 500,000 Europeans [13], so break-even needs a very small market share.

Payback: at the expected surplus of 129 JOD/month, the implementation cash outlay (456.5 JOD) is recovered in about 3.5 months (during month 4). Recovering the full budget including in-kind work (3,206.5 JOD) would take about 24.9 months from Trip Pass alone, or about 9.5 months if partner revenue reaches the expected level (≈ 210 JOD/month from 40 bookings and 2 licenses, raising the surplus to about 339 JOD/month). This assumes the expected case is reached in the first month; gradual growth extends the period.

Financial risks and responses:

Risk	Potential impact	Response
Fewer than 80 paid trips per month	Continuing operating deficit	Validate willingness to pay before launch; cut paid marketing first

Regional events reduce tourist arrivals	Lower demand	Keep marketing spend flexible; lean on hostel and partner channels
Prize or grant not received	Funding gap remains uncovered	No spending commitments against unconfirmed funding
Development takes more hours	Implementation budget overrun	Fix the v1 scope; defer non-essential features
Higher AI, hosting or maps costs	Higher fixed or usage costs	Cache results, rate-limit requests, switch AI provider or model (adapter), fall back to OSRM
Outdated transport data	Wrong recommendations and loss of trust	Last-verified dates, monthly review, in-app error reports

13. Future Plans and Expansion

Horizon	Proposed Development	Start Condition
Short term (months 1–3 after launch)	Languages (Arabic, German, French), improved offline mode, data expanded to 20 destinations	Evidence that users actively use the Reality Check and at least 1,200 plans/month
Medium term (months 4–12)	Direct booking with transport partners and in-trip companion mode	Operating break-even (80+ paid trips/month) and 5 interested transport partners
Long term (months 13–24)	Partnerships with official tourism bodies; expansion to nearby destinations	Three consecutive months of operating surplus and confirmed expansion funding

Future Resources and Cost

Future costs below are additional to the initial budget and will be added to it, with the cash requirement and break-even recalculated, once approved.

Expansion	Required resources	One-time JOD	Extra monthly JOD
Languages and offline improvements	80 development hours × 5 JOD	400	0
Direct booking with transport partners	120 development hours × 5 JOD	600	15 (payment and tools)
In-trip companion mode	150 development hours × 5 JOD	750	20 (AI, maps usage and notifications)
Expansion to nearby destinations	Launch campaign and extra support	500	100 (marketing and support)

Success Indicators

Indicator	Target	Measurement
Checked plans	1,200 per month by the end of month 6	Database records; monthly
Paid trips	At least 80 (break-even); 120 expected	Payment records; monthly
Paid conversion	At least 10% of checked plans	Paid trips / checked plans;

		monthly
Executable plans after correction	At least 90%	Manual review of a 20-plan sample; monthly
Time to a corrected plan	Under 5 minutes	Product analytics; weekly
Customer acquisition cost	No more than 1.25 JOD per paid trip	Marketing cost / new paid trips; monthly
User satisfaction	4/5 or higher	Post-trip survey; monthly
Data freshness	95% of core data verified within 90 days	Automated report; monthly
Implementation budget variance	No more than 10% of base	Actual vs. budgeted spend; during implementation

Review and Decision-Making

- The team reviews indicators monthly, with an extended review every 3 months.
- **Continue:** if 70% of targets are met, at least 90% of plans are executable and paid trips are trending towards 80 per month.
- **Adjust:** if usage is good but payments are weak (change price or add partner revenue), or if data accuracy is below target (reduce destinations and deepen verification).
- **Postpone expansion:** if break-even is not reached within 9 months, reassess pricing, costs and acquisition before approving any expansion.

Assumption Register

Implementation: 12 weeks; 2,915 JOD base cost plus 10% contingency; team work (2,750 JOD) is in-kind; no hardware purchase is included.

Operations: 255 JOD fixed costs per month plus 0.30 JOD per paid trip (payment fee + AI parsing); higher volumes may need more support or infrastructure.

Revenue: 3.5 JOD per paid trip, collected in full with no refunds; paid-trip counts are scenario assumptions, not demonstrated demand. Partner commissions and B2B licenses are excluded from the scenarios.

Funding: 500 JOD cash and 2,750 JOD in-kind are treated as available; any prize or grant is excluded until confirmed.

Scope of results: operating results exclude implementation cost recovery, taxes and unlisted expenses; the six-month funding calculation excludes sales receipts.

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Source 3

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Accessed 24/09/2026 — free plan limits and Pro plan price.

Source 12

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Source 13

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Source 14

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<https://www.jett.com.jo>

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Source 15

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Source 16

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<https://play.google.com/store/apps/details?id=com.imagine.jordanpass.superapplication> — <https://apps.apple.com/us/app/visit-jordan/id6745452007>

Accessed 24/09/2026 — 500+ downloads, features, release 5 Aug 2025; no itinerary planner or feasibility check.

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Source 19

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Source 20

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<https://platform.claude.com/docs/en/about-claude/pricing>

Accessed 24/09/2026 — small-model rates of about USD 1 per million input tokens and USD 5 per million output tokens, used to budget AI parsing.

Source 21

Open-Meteo, "Free Weather API", 2026.

<https://open-meteo.com/>

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baseline provider for the 7-day forecast.